



Analysing E-Learning Platform Purchases

Using MySQL



Analysing E-Learning Platform Purchases using MySQL

Summary:

This project focuses on analysing course purchase data from an e-learning platform using MySQL. The goal is to understand sales performance across different course categories and learner behaviour patterns. The analysis helps identify top-performing courses, revenue-driving categories, and purchasing trends to support data-driven business decisions.

Tools & Technologies:

Database Creation: E Learning Platform

- SQL Queries for Data Analysis, Aggregations, Grouping, Filtering and Sorting.

Analysis Workflow:

- Data Exploration - Analysed course purchase records, examined categories, learners, quantities, and revenue.
- Performance Analysis - Identified top-performing categories, evaluated best-selling courses. Measured purchase volumes and revenue contribution.
- Learner Behaviour Analysis - Analysed cross-category purchases observed multi-course buying patterns.
- Optimization Insights - Detected underperforming courses with zero purchases compared learner engagement across categories.

Key Insights:

- ✓ A small number of categories contribute the majority of revenue.
- ✓ Top 3 courses generate a significant portion of total sales.
- ✓ Many learners purchase across multiple categories, showing cross-selling opportunities.
- ✓ Some courses recorded zero purchases, indicating improvement areas.
- ✓ Higher revenue categories tend to have more unique learners.

Recommendations:

- Focus marketing efforts on high-performing categories.
- Promote best-selling courses as flagship offerings.
- Create bundle offers to encourage cross-category purchases.
- Improve or redesign underperforming courses.
- Increase engagement through discounts, trials, or updated content.
- Expand global outreach to increase international sales.

Conclusion:

This project demonstrates how MySQL-based data analysis can uncover valuable business insights from e-learning purchase data. By leveraging SQL queries and structured analysis, organizations can optimize course offerings, improve learner engagement, and drive revenue growth.

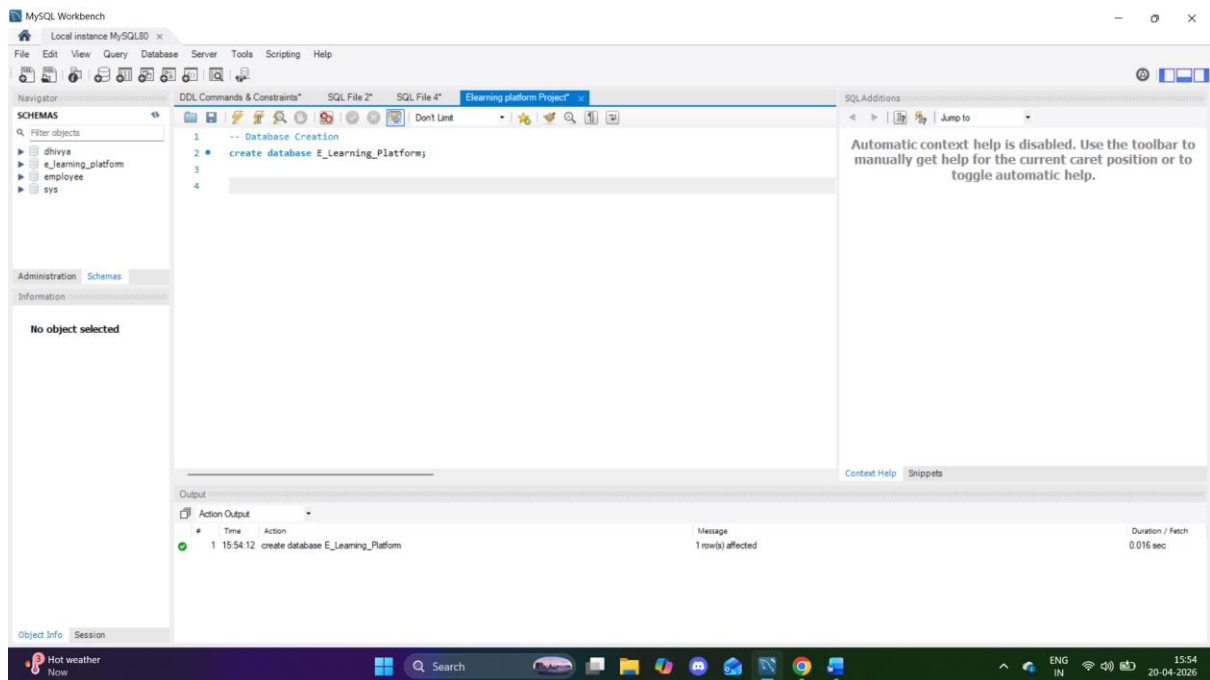
Project Documentation

SQL Output

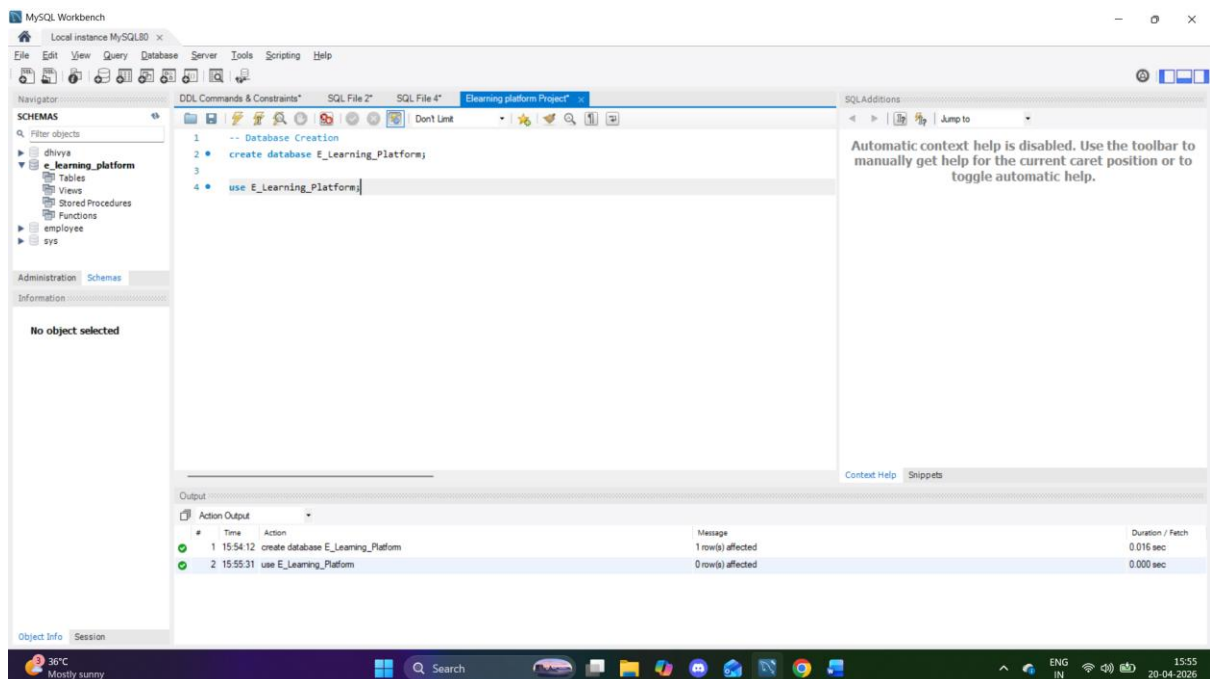
Screenshot

1. Database Creation:

The database **E_Learning_Platform** was successfully created.

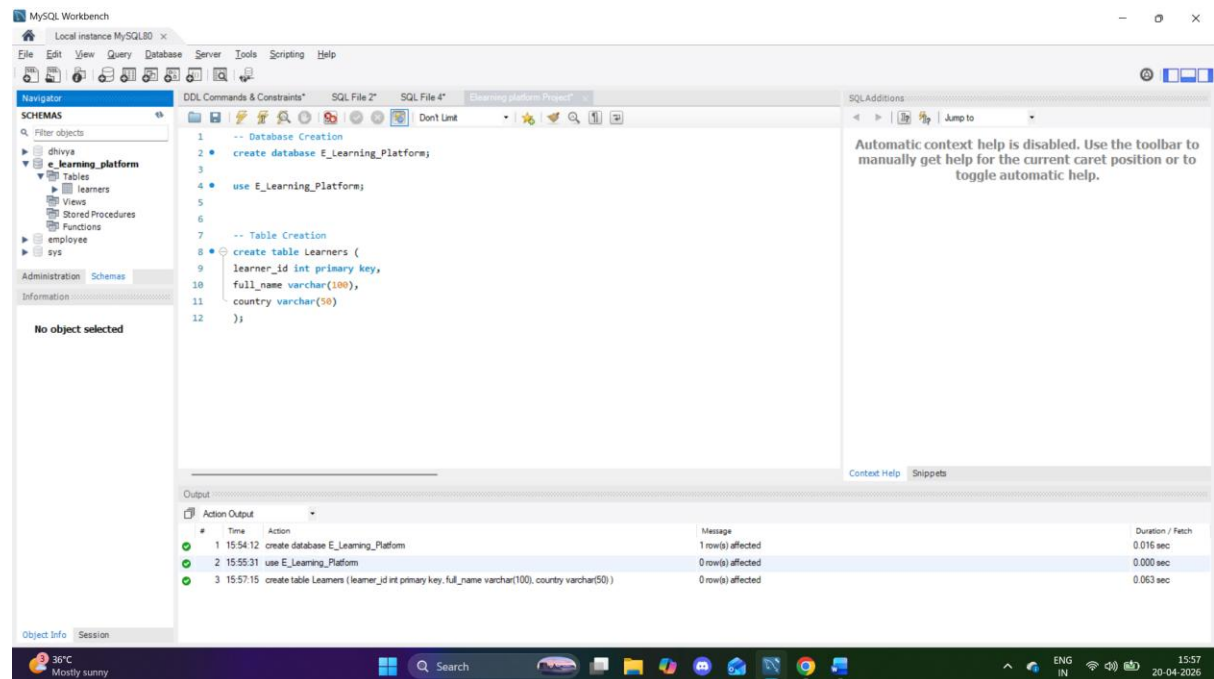


Using the USE command to perform further operations.



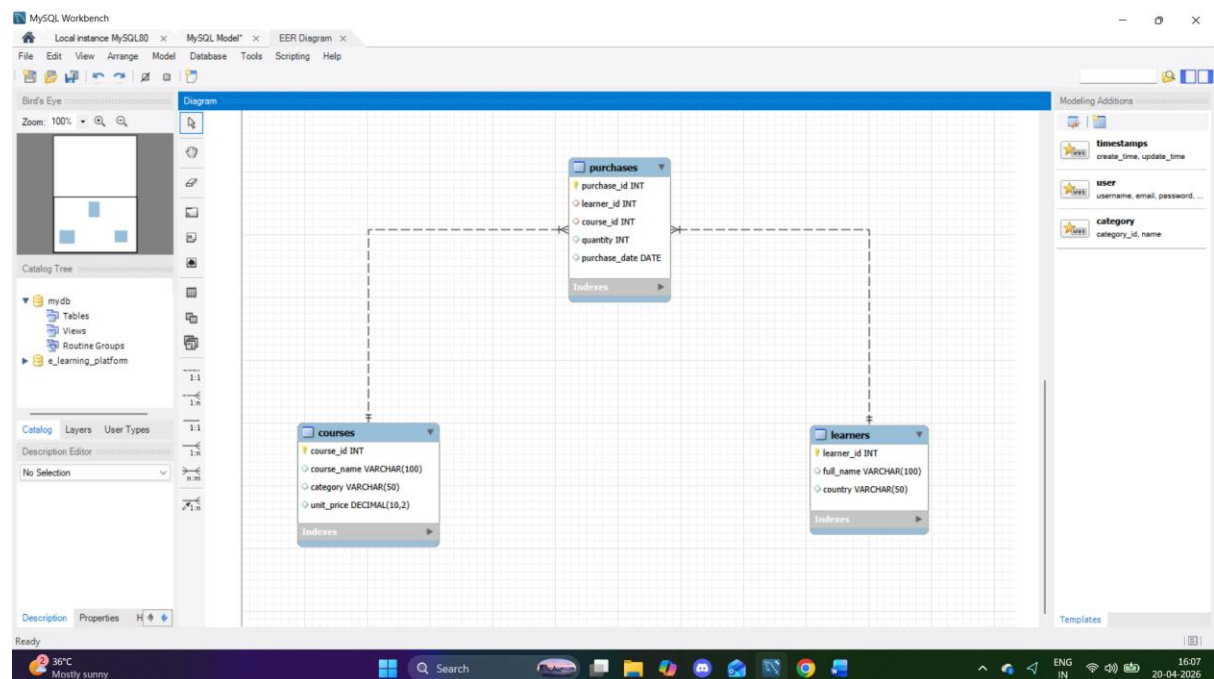
2. Table Creation:

Three main tables were created Learners, Courses and Purchases. For eg., I shared Learners Table screenshot.



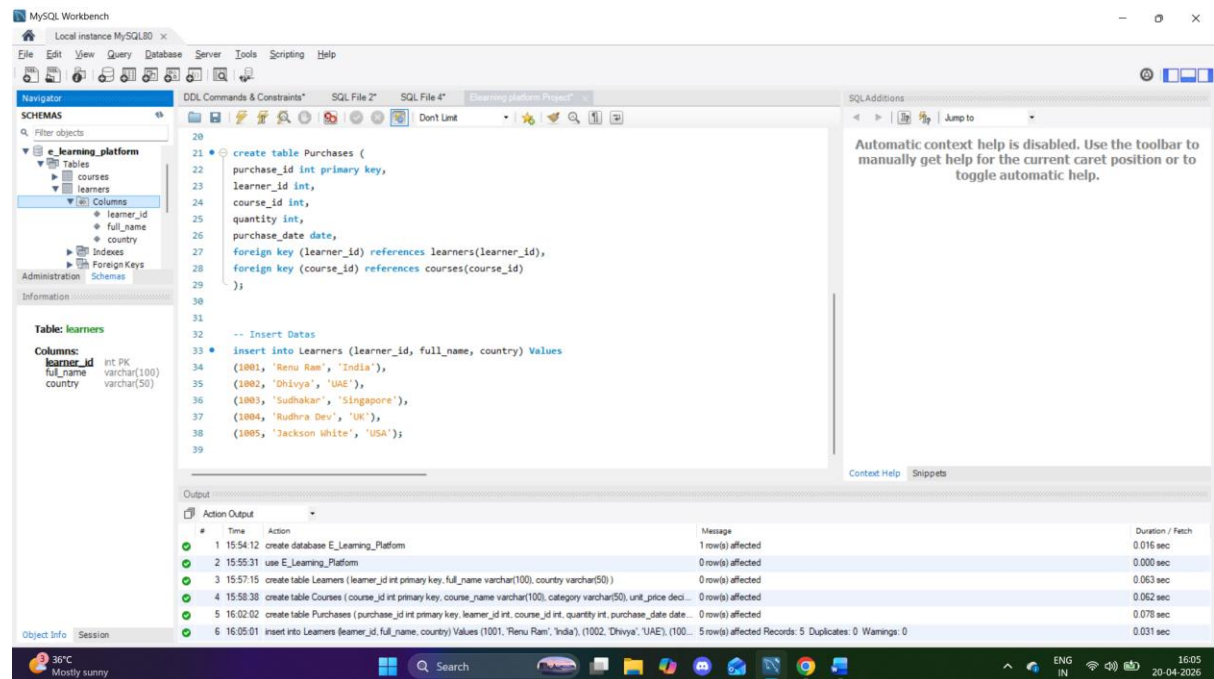
3. Create all three tables in MySQL with appropriate data types and relationships.

Database → Reverse Engineer (It auto-generates ER diagram)



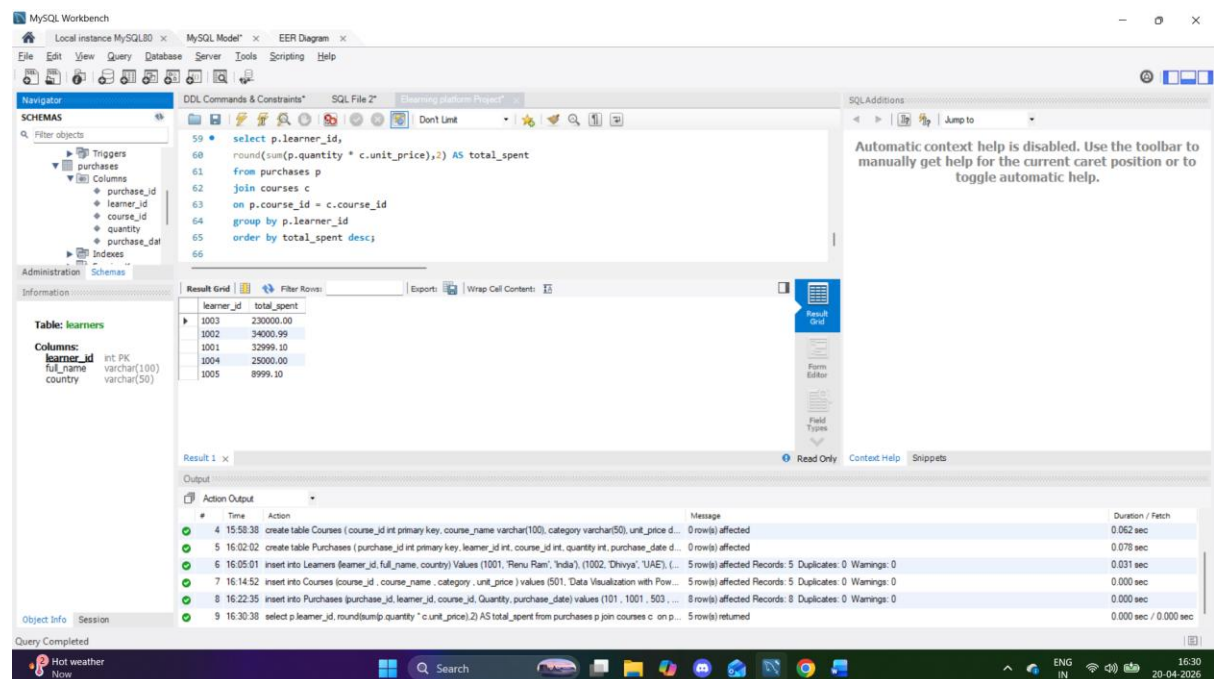
4. Insert sample datas:

Sample records were successfully inserted into all three tables using INSERT INTO statements.



5. Data Exploration:

Basic queries were performed



6. Using Joins:

Different types of joins were used to combine data from multiple tables, and it helps to analyzing the relationships between tables.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
67 select l.full_name as learner_name,
68 c.course_name, c.category, p.quantity,
69 round(p.quantity * c.unit_price, 2) as total_amount,
70 p.purchase_date from purchases p
71 inner join learners l on p.learner_id = l.learner_id
72 inner join courses c on p.course_id = c.course_id
73 order by total_amount asc;
```

The Result Grid displays the following data:

learner_name	course_name	category	quantity	total_amount	purchase_date
Renu Ram	Introduction of Data Science	Data Science	1	8999.10	2026-01-08
Jackson White	Introduction of Data Science	Data Science	1	8999.10	2026-03-10
Ohviya	Digital Marketing Strategy	Marketing	1	12000.00	2026-04-20
Ohviya	Cybersecurity Basics	Information Security	1	22000.99	2026-02-05
Renu Ram	Digital Marketing Strategy	Marketing	2	24000.00	2026-03-29
Rudhra Dev	Data Visualization with Power BI	Business Intelligence	1	25000.00	2026-02-16
Sudhakar	Data Visualization with Power BI	Business Intelligence	2	50000.00	2026-01-30
Sudhakar	Python for Data Analysis	Programming	3	180000.00	2026-04-12

The Action Output shows the execution of the query, indicating that 8 rows were returned.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
81
82 select l.full_name as learner_name, c.course_name, c.category, p.quantity,
83 round(p.quantity * c.unit_price, 2) as total_amount,
84 p.purchase_date from Purchases p
85 right join Courses c on p.course_id = c.course_id
86 left join Learners l on p.learner_id = l.learner_id
87 order by course_name;
```

The Result Grid displays the following data:

learner_name	course_name	category	quantity	total_amount	purchase_date
Ohviya	Cybersecurity Basics	Information Security	1	22000.99	2026-02-05
Sudhakar	Data Visualization with Power BI	Business Intelligence	2	50000.00	2026-01-30
Rudhra Dev	Data Visualization with Power BI	Business Intelligence	1	25000.00	2026-02-16
Renu Ram	Digital Marketing Strategy	Marketing	2	24000.00	2026-03-29
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learner_name	course_name	category	quantity	total_amount	purchase_date
Dhivya	Cybersecurity Basics	Information Security	1	22000.99	2026-02-05
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Jackson White	Introduction of Data Science	Data Science	1	8999.10	2026-03-10
Sudhakar	Python for Data Analysis	Programming	3	180000.00	2026-04-12

The Output pane shows the execution log with the following messages:

```

# Time Action Message Duration / Fetch
8 16:22:35 Insert into Purchases (purchase_id, learner_id, course_id, quantity, purchase_date) values (101, 1001, 500, ... 8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0 0.000 sec / 0.000 sec
9 16:30:38 select p.learner_id, round(sum(p.quantity * c.unit_price), 2) AS total_spent from purchases p join courses c on p.p... 5 row(s) returned 0.000 sec / 0.000 sec
10 16:36:36 select l.full_name as learner_name, c.course_name, c.category, p.quantity, round(p.quantity * c.unit_price, 2) ... 8 row(s) returned 0.000 sec / 0.000 sec
11 16:41:12 select l.full_name as learner_name, c.course_name, c.category, p.quantity, round(p.quantity * c.unit_price, 2) ... 8 row(s) returned 0.016 sec / 0.000 sec
12 16:44:25 select l.full_name as learner_name, c.course_name, c.category, p.quantity, round(p.quantity * c.unit_price, 2) a... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec
13 16:44:38 select l.full_name as learner_name, c.course_name, c.category, p.quantity, round(p.quantity * c.unit_price, 2) a... 8 row(s) returned 0.000 sec / 0.000 sec
14 16:44:38 select l.full_name as learner_name, c.course_name, c.category, p.quantity, round(p.quantity * c.unit_price, 2) a... 8 row(s) returned 0.000 sec / 0.000 sec

```

7. Analytical Queries :

Q1. Display each learner's total spending (quantity × unit_price) along with their country.

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```

95 inner join purchases p
96 on l.learner_id = p.learner_id
97 inner join courses c
98 on p.course_id = c.course_id
99 group by l.learner_id, l.full_name, l.country
100 order by total_spent;
101

```

The Results Grid shows the following data:

learner_id	full_name	country	total_spent
1005	Jackson White	USA	8999.10
1004	Rudra Dev	UK	25000.00
1001	Renu Ram	India	32999.10
1002	Dhivya	UAE	34000.99
1003	Sudhakar	Singapore	230000.00

The Output pane shows the execution log with the following messages:

```

# Time Action Message Duration / Fetch
9 16:30:38 select p.learner_id, round(sum(p.quantity * c.unit_price), 2) AS total_spent from purchases p join courses c on p.p... 5 row(s) returned 0.000 sec / 0.000 sec
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13 16:44:38 select l.full_name as learner_name, c.course_name, c.category, p.quantity, round(p.quantity * c.unit_price, 2) a... 8 row(s) returned 0.000 sec / 0.000 sec
14 16:44:38 select l.learner_id, l.full_name, l.country, round(sum(p.quantity * c.unit_price), 2) as total_spent from learners l in... 5 row(s) returned 0.000 sec / 0.000 sec

```

Q2. Find the top 3 most purchased courses based on total quantity sold.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

103 sum(p.quantity) as total_quantity_sold
104 from courses c
105 inner join purchases p
106 on p.course_id = c.course_id
107 group by c.course_id, c.course_name, c.category
108 order by total_quantity_sold desc
109 limit 3;
110

```

The Result Grid shows the following data:

course_id	course_name	category	total_quantity_sold
501	Data Visualization with Power BI	Business Intelligence	3
502	Python for Data Analysis	Programming	3
504	Digital Marketing Strategy	Marketing	3

The Action Output pane shows the execution of the query, with a message indicating that the query was successful and returned 3 rows.

Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

112 round(sum(p.quantity * c.unit_price), 2) as total_revenue,
113 count(distinct p.learner_id) as unique_learners
114 from courses c
115 inner join purchases p
116 on p.course_id = c.course_id
117 group by c.category
118 order by total_revenue;
119

```

The Result Grid shows the following data:

category	total_revenue	unique_learners
Data Science	17998.20	2
Information Security	22000.99	1
Marketing	36000.00	2
Business Intelligence	75000.00	2
Programming	180000.00	1

The Action Output pane shows the execution of the query, with a message indicating that the query was successful and returned 5 rows.

Q4. List all learners who have purchased courses from more than one category.

MySQL Workbench interface showing the query execution for Q4. The query is:

```

122 FROM learners l
123 INNER JOIN purchases p
124 ON p.learner_id = l.learner_id
125 INNER JOIN courses c
126 ON p.course_id = c.course_id
127 GROUP BY l.learner_id, l.full_name
128 HAVING COUNT(DISTINCT c.category) > 1
129

```

The result grid shows the following data:

learner_id	learner_name	category_count
1001	Renu Ram	2
1002	Divya	2
1003	Sudhakar	2

Q5. Identify courses that have not been purchased at all.

MySQL Workbench interface showing the query execution for Q5. The query is:

```

128 HAVING COUNT(DISTINCT c.category) > 1;
129
130 SELECT c.course_id, c.course_name, c.category
131 FROM courses c
132 LEFT JOIN purchases p
133 ON c.course_id = p.course_id
134 WHERE p.course_id = null;
135

```

The result grid shows 0 rows, indicating that no courses have been purchased at all.

--- Thank you ---